

Orthogonal Matrices

A square matrix A over the field R is called Orthogonal if $A^t A = I = A A^t$

OR $A^t = A^{-1}$

From the def it follows that

A^t is inverse of A
 $\therefore A^t A = I$

- A is orthogonal iff A^{-1} is orthogonal
- If A is orthogonal then A is symmetric iff $A^2 = I$
- If A & B are orthogonal then A^t and B^t are also orthogonal.
- If A is orthogonal then the col vectors are unit length and their dot products are zero i.e. Columns of A form an orthonormal set.
- If A is orthogonal then the rows of A form an orthonormal set.
- A is orthogonal iff A^t is orthogonal.

Theorem The following conditions for a square Matrix A are equivalent:

- A is orthogonal
- The rows of A form an orthonormal set.
- The col of A form an orthonormal set.

Proof Let $A = [a_{ij}]_{n \times n}$ be an Orthogonal Matrix i.e. $AA^t = I = A^t A$

Let R_i & C_j denote the i^{th} row & j^{th} col of A where $1 \leq i \leq n$

Let $C_{ij} = (i, j)$ element of AA^t

= Sum of products of the corresponding elements in the i^{th} row of A and j^{th} col of A^t .

= Sum of products of the corresponding elements in the i^{th} row of A and j^{th} row of A

$$= a_{i1}a_{j1} + a_{i2}a_{j2} + \dots + a_{in}a_{jn}$$

$$C_{ij} = \langle R_i, R_j \rangle \quad \text{--- (1)}$$

Matrix $[C_{ij}] = AA^t$ (I.P. of two rows)

$$[C_{ij}] = I$$

$$\therefore \langle R_i, R_j \rangle = C_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

$$\therefore C_{ij} = \langle R_i, R_j \rangle$$

$$\therefore \langle R_i, R_j \rangle = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases}$$

$\Rightarrow$ Rows of A are orthonormal

$\Rightarrow$ (i) $\Rightarrow$ (ii) (from statement & theorem)

i^{th} row of $A = (a_{i1}, a_{i2}, \dots, a_{in})$
 j^{th} row of $A = (a_{j1}, a_{j2}, \dots, a_{jn})$
for detail see examples

$\therefore A$ is orthogonal so $AA^t = I$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{pmatrix}$$

when $i=j$ $C_{ij}=1$
when $i \neq j$ $C_{ij}=0$

Conversely if rows of A form an orthonormal set, then

$$\langle R_i, R_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\therefore \langle R_i, R_j \rangle = C_{ij} \quad \therefore C_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\Rightarrow [C_{ij}] = I$$

$$\therefore AA^t = [C_{ij}] \quad \therefore AA^t = I \Rightarrow A \text{ is orthogonal Matrix}$$

So (i) $\Rightarrow$ (i') Hence (i) $\Leftrightarrow$ (ii)

Now Consider the Matrix $A^t A = [d_{ij}]$

d_{ij} = (i,j) element of $A^t A$
= Sum of the products of the corresponding elements in i^{th} row of A^t and j^{th} col of A .

= Sum of the products of the corresponding elements in i^{th} col of A and j^{th} col of A

$$= a_{1i}a_{1j} + a_{2i}a_{2j} + \dots + a_{ni}a_{nj}$$

= Inner product of i^{th} & j^{th} col matrices

$$\begin{aligned} i^{\text{th}} \text{ row of } A^t &= (a_{1i} \ a_{2i} \ \dots \ a_{ni}) \\ i^{\text{th}} \text{ col of } A &= \begin{pmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \end{pmatrix} \\ j^{\text{th}} \text{ col of } A &= \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{pmatrix} \end{aligned}$$

$$d_{ij} = \langle C_i, C_j \rangle$$

$$[d_{ij}] = A^t A = I$$

$\therefore A$ is orthogonal Matrix

$$\Rightarrow d_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\therefore d_{ij} = \langle C_i, C_j \rangle$$

$$\therefore \langle C_i, C_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$\Rightarrow$ cols C_i & C_j are orthonormal

$\Rightarrow$ (i) $\Rightarrow$ (iii)

Conversely if columns of A form orthonormal set, then

$$\langle C_i, C_j \rangle = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\therefore \langle C_i, C_j \rangle = d_{ij} \quad \therefore d_{ij} = \begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$\Rightarrow [d_{ij}] = I$$

$$\therefore [d_{ij}] = A^t A \quad \therefore A^t A = I \Rightarrow A \text{ is orthogonal Matrix}$$

So (iii) $\Rightarrow$ (i) Hence (i) $\Leftrightarrow$ (iii)

Now (i) $\Leftrightarrow$ (ii) & (ii) $\Leftrightarrow$ (iii) $\Rightarrow$ (i) $\Leftrightarrow$ (iii)

NOTE (ii) $\Leftrightarrow$ (iii) so A is orthogonal iff A^t is orthogonal

Example 13 Show that the rows (columns) of the matrix

$$A = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ form an orthonormal set.}$$

Sol To show that the rows (columns) of the matrix A form an orthonormal set we have only to show that A is orthogonal (Th 7.12 on page 277)
(If A is orthogonal then $AA^t = I$)

$$\begin{aligned} \therefore AA^t &= \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta + 0 & \cos \theta \sin \theta - \sin \theta \cos \theta + 0 & 0 + 0 + 0 \\ \sin \theta \cos \theta - \cos \theta \sin \theta + 0 & \sin^2 \theta + \cos^2 \theta + 0 & 0 + 0 + 0 \\ 0 + 0 + 0 & 0 + 0 + 0 & 0 + 0 + 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

Thus A is orthogonal
So Rows of A form an orthonormal set.

Example 14 Find an orthogonal matrix A whose first row is $\left[\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

Sol First Method Let $R_1 = \left[\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right]$

A vector (row) orthogonal to R_1 is $\star R_2 = \begin{vmatrix} e_1 & e_2 \\ \frac{1}{3} & \frac{2}{3} \end{vmatrix} + \begin{vmatrix} e_2 & e_3 \\ \frac{2}{3} & \frac{2}{3} \end{vmatrix} + \begin{vmatrix} e_3 & e_1 \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix}$

$$\begin{aligned} &= e_1 \left(\frac{2}{3}\right) - e_2 \left(\frac{1}{3}\right) + e_2 \left(\frac{2}{3}\right) - e_3 \left(\frac{2}{3}\right) + e_3 \left(\frac{1}{3}\right) - e_1 \left(\frac{2}{3}\right) \\ &= e_2 \left(\frac{2}{3} - \frac{1}{3}\right) + e_3 \left(-\frac{2}{3} + \frac{1}{3}\right) = e_2 \left(\frac{1}{3}\right) + e_3 \left(-\frac{1}{3}\right) \\ \star R_2 &= \left(0, \frac{1}{3}, -\frac{1}{3}\right) \end{aligned}$$

$$R_2 = \frac{\star R_2}{\|\star R_2\|} = \frac{(0, \frac{1}{3}, -\frac{1}{3})}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}} (0, 1, -1) = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

A vector (row) orthogonal to R_1 & $\star R_2$ is $\star R_3 = \begin{vmatrix} e_1 & e_2 & e_3 \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{3} & -\frac{1}{3} \end{vmatrix} = e_1 \left(-\frac{2}{9} - \frac{2}{9}\right) - e_2 \left(-\frac{1}{9}\right) + e_3 \left(\frac{1}{9}\right)$

$$\begin{aligned} &= -\frac{4}{9} e_1 + \frac{1}{9} e_2 + \frac{1}{9} e_3 \\ &= \left(-\frac{4}{9}, \frac{1}{9}, \frac{1}{9}\right) \end{aligned}$$

$$R_3 = \frac{\star R_3}{\|\star R_3\|} = \frac{(-\frac{4}{9}, \frac{1}{9}, \frac{1}{9})}{\frac{1}{9}} = (-4, 1, 1) = \frac{1}{3\sqrt{2}} \left(-\frac{4}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \left(-\frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}\right)$$

$$\therefore \text{Orthogonal Matrix } A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} \end{bmatrix}$$

To check Find $A^t A$.
we get $A^t A = I$

Example 14 2nd Method

7.2A-4

Q4

$$\text{Let } R_1 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right)$$

A vector orthogonal to R_1 is R_2^*

$$R_2^* = \begin{vmatrix} e_1 & e_2 \\ \frac{1}{3} & \frac{2}{3} \end{vmatrix} + \begin{vmatrix} e_2 & e_3 \\ \frac{2}{3} & \frac{2}{3} \end{vmatrix} + \begin{vmatrix} e_3 & e_1 \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix}$$

$$= e_1 \left(\frac{2}{3} \right) - e_2 \left(\frac{1}{3} \right) + e_2 \left(\frac{2}{3} \right) - e_3 \left(\frac{2}{3} \right) + e_3 \left(\frac{1}{3} \right) - e_1 \left(\frac{2}{3} \right)$$

$$R_2^* = e_2 \left(\frac{1}{3} \right) + e_3 \left(-\frac{1}{3} \right) = \left(0, \frac{1}{3}, -\frac{1}{3} \right)$$

$$\|R_2^*\| = \sqrt{0^2 + \frac{1}{9} + \frac{1}{9}} = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3}$$

$$R_2 = \frac{R_2^*}{\|R_2^*\|} = \frac{\left(0, \frac{1}{3}, -\frac{1}{3} \right)}{\frac{\sqrt{2}}{3}} = \frac{3}{\sqrt{2}} \left(0, \frac{1}{3}, -\frac{1}{3} \right) = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

A vector orthogonal to R_1 & R_2 is R_3^*

$$R_3^* = \begin{vmatrix} e_1 & e_2 & e_3 \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{vmatrix} = e_1 \left(-\frac{2}{3\sqrt{2}} - \frac{2}{3\sqrt{2}} \right) - e_2 \left(-\frac{1}{3\sqrt{2}} \right) + e_3 \left(\frac{1}{3\sqrt{2}} \right)$$

$$= \left(-\frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}} \right)$$

$$R_3 = \frac{R_3^*}{\|R_3^*\|} = \frac{\left(-\frac{4}{3\sqrt{2}}, \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}} \right)}{\frac{1}{\sqrt{2}}} = \left(-\frac{4}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

$$\text{Hence orthogonal Matrix } A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} \end{bmatrix}$$

3rd Method

$$\text{Let } v_1 = \left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3} \right) \text{ \& } \omega_2 = (x, y, z)$$

$$\text{Let } \omega_2 \text{ is orthogonal to } v_1 \therefore \langle v_1, \omega_2 \rangle = 0$$

$$\Rightarrow \frac{x}{3} + \frac{2y}{3} + \frac{2z}{3} = 0 \text{ or } x + 2y + 2z = 0 \text{ --- (I)}$$

$$\text{If we take } x=0 \text{ then } 2y + 2z = 0 \Rightarrow z = -y$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ y \\ -y \end{bmatrix} = y \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \therefore \omega_2 = (x, y, z) = (0, 1, -1)$$

$$v_2 = \frac{\omega_2}{\|\omega_2\|} = \frac{(0, 1, -1)}{\sqrt{2}} = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right)$$

$$\text{Now let } \omega_3 = (x, y, z) \text{ be orthogonal to } v_1 \text{ \& } v_2$$

$$\therefore \langle v_1, \omega_3 \rangle = 0 \Rightarrow \frac{x}{3} + \frac{2y}{3} + \frac{2z}{3} = 0 \Rightarrow x + 2y + 2z = 0 \text{ --- (II)}$$

$$\langle v_2, \omega_3 \rangle = 0 \Rightarrow \frac{y}{\sqrt{2}} - \frac{z}{\sqrt{2}} = 0 \Rightarrow y - z = 0 \Rightarrow y = z \text{ Put in (II)}$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -4z \\ z \\ z \end{bmatrix} = z \begin{bmatrix} -4 \\ 1 \\ 1 \end{bmatrix} \therefore \omega_3 = (-4, 1, 1)$$

$$v_3 = \frac{\omega_3}{\|\omega_3\|} = \frac{(-4, 1, 1)}{\sqrt{18}} = \left(-\frac{2\sqrt{2}}{3}, \frac{1}{3\sqrt{2}}, \frac{1}{3\sqrt{2}} \right) \text{ Hence } A = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{2\sqrt{2}}{3} & \frac{1}{3\sqrt{2}} & \frac{1}{3\sqrt{2}} \end{bmatrix}$$

Note
We can check our answer
by $\langle R_1, R_2 \rangle = 0 = \langle R_1, R_3 \rangle$
 $0 = \langle R_2, R_3 \rangle$
 $\langle R_1, R_1 \rangle = \langle R_2, R_2 \rangle = \langle R_3, R_3 \rangle = 1$

$$\text{OR } AA^t = I$$

$\checkmark \Rightarrow$ order of A is 3×3
"A must be square.
 v_1 is first row of A
 v_2 is 2nd row of A
(we may take $y=0$ or $z=0$)

Ex 7.2
Q1 If A is an orthogonal matrix show that $A = 1$ or -1

Sol Since A is orthogonal Matrix

$$\therefore AA^t = I$$

$$\det(AA^t) = \det I$$

By Product Theorem

$$\Rightarrow \det(A) \cdot \det(A^t) = 1$$

$$\therefore (\det(A) = \det A^t \cdot \det B)$$

$$\Rightarrow \det(A) \cdot \det(A) = 1$$

$$\therefore \det(A^t) = \det(A)$$

$$\Rightarrow [\det(A)]^2 = 1$$

$$\det(A) = \pm 1 \text{ Proved}$$

2nd Method

Since A is orthogonal Matrix

$$\therefore A^t = A^{-1}$$

$$\det(A^t) = \det(A^{-1})$$

$$\det A = \frac{1}{\det(A)}$$

$$\therefore \det(A^t) = \det A$$

$$[\det A]^2 = 1$$

$$\det A = \pm 1 \text{ Proved.}$$

Prove that $\det A^t = \frac{1}{\det A}$

$$\text{We know } A^t A = I$$

$$\det(A^t A) = \det I$$

$$\text{By Product Theorem } \det A^t \det A = 1$$

$$\det A^t = \frac{1}{\det A}$$

proved.

Q2 Find an orthogonal matrix whose first row is $(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$

Sol Let $R_1 = (\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$ (given)

A vector orthogonal to R_1 is

$$R_2^* = \begin{bmatrix} e_1 & e_2 \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$= \frac{2}{\sqrt{5}} e_1 - \frac{1}{\sqrt{5}} e_2$$

$$= (\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}})$$

$$\|R_2^*\| = \sqrt{\frac{4}{5} + \frac{1}{5}} = \sqrt{\frac{5}{5}} = 1$$

$$R_2 = \frac{R_2^*}{\|R_2^*\|} = (\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}) = (\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}})$$

$$\therefore A = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix}$$

To Check $AA^t = I$

$$AA^t = \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & -\frac{1}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \frac{1}{5} + \frac{4}{5} & \frac{2}{5} - \frac{2}{5} \\ \frac{2}{5} + \frac{2}{5} & \frac{4}{5} + \frac{1}{5} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ proved}$$

Q3 Find an orthogonal matrix whose first row is multiple of (1,1,1)

Sol $\omega_1 = (1, 1, 1)$ given

$$\|\omega_1\| = \sqrt{3}$$

Normalised first Row $V_1 = \frac{\omega_1}{\|\omega_1\|} = \frac{(1, 1, 1)}{\sqrt{3}} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$

Let ω_2 is orthogonal to V_1 , where $\omega_2 = (x, y, z)$

$$\therefore \langle V_1, \omega_2 \rangle = 0$$

$$\Rightarrow \frac{x}{\sqrt{3}} + \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}} = 0$$

$$\Rightarrow x + y + z = 0$$

Put $x = 0$

$$\Rightarrow y + z = 0 \Rightarrow z = -y$$

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ y \\ -y \end{bmatrix} = y \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$\therefore \omega_2 = (0, 1, -1)$$

$$\|\omega_2\| = \sqrt{0^2 + 1^2 + (-1)^2} = \sqrt{2}$$

$$V_2 = \frac{\omega_2}{\|\omega_2\|} = \frac{(0, 1, -1)}{\sqrt{2}} = \left(0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$$

A vector orthogonal to V_1 & V_2 is

$$\omega_3 = \begin{vmatrix} e_1 & e_2 & e_3 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{vmatrix}$$

$$= e_1 \left(-\frac{1}{\sqrt{6}} - \frac{1}{\sqrt{6}}\right) - e_2 \left(-\frac{1}{\sqrt{6}}\right) + e_3 \left(\frac{1}{\sqrt{6}}\right)$$

$$\omega_3 = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$\|\omega_3\| = \sqrt{\frac{4}{6} + \frac{1}{6} + \frac{1}{6}} = 1$$

$$V_3 = \frac{\omega_3}{\|\omega_3\|} = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

Hence the required orthogonal matrix is $\begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}$

we cannot use

$$\begin{vmatrix} e_1 & e_2 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix} + \begin{vmatrix} e_2 & e_3 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix} + \begin{vmatrix} e_3 & e_1 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{vmatrix}$$

because it becomes 0, and when it becomes 0 it is not applicable.

Q4 Find an orthogonal matrix whose first row is $(0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}})$

$$\text{Let } R_1 = \left(0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$$

A vector (row) orthogonal to R_1 is $R_2^* = \begin{vmatrix} e_1 & e_2 \\ 0 & \frac{1}{\sqrt{5}} \end{vmatrix} + \begin{vmatrix} e_2 & e_3 \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \end{vmatrix} + \begin{vmatrix} e_3 & e_1 \\ \frac{2}{\sqrt{5}} & 0 \end{vmatrix}$

$$= e_1\left(\frac{1}{\sqrt{5}}\right) + \frac{2}{\sqrt{5}}e_2 - \frac{1}{\sqrt{5}}e_3 + (0 - \frac{2}{\sqrt{5}}e_1)$$

$$= e_1\left(\frac{1}{\sqrt{5}} - \frac{2}{\sqrt{5}}\right) + \frac{2}{\sqrt{5}}e_2 - \frac{1}{\sqrt{5}}e_3$$

$$R_2^* = -\frac{1}{\sqrt{5}}e_1 + \frac{2}{\sqrt{5}}e_2 - \frac{1}{\sqrt{5}}e_3 = \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right)$$

$$\|R_2^*\| = \sqrt{\frac{1}{5} + \frac{4}{5} + \frac{1}{5}} = \sqrt{\frac{6}{5}}$$

$$R_2 = \frac{R_2^*}{\|R_2^*\|} = \frac{\left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right)}{\sqrt{\frac{6}{5}}} = \frac{\sqrt{5}}{\sqrt{6}} \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right) = \left(-\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}\right)$$

A vector orthogonal to R_1 & R_2 is $R_3^* = \begin{vmatrix} e_1 & e_2 & e_3 \\ 0 & \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{vmatrix}$

$$= e_1\left(\frac{1}{\sqrt{5}} - \frac{4}{\sqrt{30}}\right) - e_2\left(\frac{2}{\sqrt{30}}\right) - e_3\left(\frac{1}{\sqrt{30}}\right)$$

$$R_3^* = \left(-\frac{5}{\sqrt{30}}, -\frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}\right)$$

$$\|R_3^*\| = \sqrt{\frac{25}{30} + \frac{4}{30} + \frac{1}{30}} = 1$$

$$R_3 = \frac{R_3^*}{\|R_3^*\|} = \left(-\frac{5}{\sqrt{30}}, -\frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}\right)$$

Hence orthogonal Matrix $A = \begin{pmatrix} 0 & \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ -\frac{5}{\sqrt{30}} & \frac{2}{\sqrt{30}} & \frac{1}{\sqrt{30}} \end{pmatrix}$

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Q5 Show that the products and inverses of orthogonal matrices are orthogonal. Hence show that orthogonal matrices form a group under multiplication.

Sol Let A & B be orthogonal matrices. So $A^t A = I = A^t = A^{-1}$
 $B^t B = I \Rightarrow B^t = B^{-1}$

Now To Prove Product of orthogonal Matrix is Orthogonal, i.e. (AB) is orthogonal.

Therefore $(AB)^t (AB) = B^t A^t AB$ we show $(AB)^t (AB) = I$
 $= B^t (A^t A) B$
 $= B^t I B$
 $= B^t B$
 $(AB)^t (AB) = I$

So AB is an orthogonal Matrix.

Now To Prove Inverse of orthogonal Matrix is orthogonal, i.e. A^{-1} is orthogonal.

Therefore $(A^{-1})^t (A^{-1}) = (A^{-1})^t (A^t)$ we show $(A^{-1})^t (A^{-1}) = I$
 $= (AA^{-1})^t$ $(A^{-1})^t = A^t$ since A is orthogonal
 $= (I)^t$ $\therefore (AB)^t = B^t A^t$
 $(A^{-1})^t (A^{-1}) = I$

Hence A^{-1} is orthogonal.

To Prove Set of orthogonal matrices form a group under multiplication.

Let $\mathcal{G}$ = Set of orthogonal matrices.

i) Let $A, B \in \mathcal{G}$ then $AB \in \mathcal{G}$ $\because$ product of orthogonal is orthogonal.

Hence $\mathcal{G}$ is closed under B.O of multiplication.

ii) In Matrices Associative law holds.

iii) Identity Matrix is always orthogonal Matrix. So $I \in \mathcal{G}$.

iv) Inverse of each orthogonal matrix is orthogonal (as solved above).

So Inverse of each orthogonal matrix belonging to $\mathcal{G}$ is in $\mathcal{G}$.

So Inverse exists in $\mathcal{G}$.

All conditions are satisfied, Hence $\mathcal{G}$ is a group under multiplication.